

STACK MCQ XML Generator

Windows Setup Guide

From Git clone and first-time setup to opening localhost:4173 in your browser

Audience For users starting from a fresh Git clone and opening the local web app for the first time.

What you will accomplish

- Install the required software on Windows.
- Clone stack_questions from GitHub.
- Configure STACK code and enable local Maxima evaluation.
- Start the local web app and open it in a browser.

Note An internet connection is required for the initial setup. No additional Python packages are required.

Quick path

1	Install Git, Python, and Maxima.
2	Clone stack_questions in PowerShell.
3	Double-click start-mcq-webapp.bat.
4	Press Enter at the first prompt to download STACK automatically.
5	Open http://127.0.0.1:4173/ in your browser.

1. Install the required software

1 Install Git for Windows

Download Git for Windows from the official website and install it using the standard options.

<https://git-scm.com/download/win>

2 Install Python 3.10 or later

Download the Windows installer from Python.org. On the first installer screen, enable **Add python.exe to PATH**.

<https://www.python.org/downloads/windows/>

3 Install Maxima

Download and run the Windows win64.exe installer from the official Maxima download page.

<https://maxima.sourceforge.io/download.html>

Important

After installation, close PowerShell and open it again so that the updated PATH is available.

2. Clone stack_questions

Open PowerShell from the Start menu. The following example uses your Documents folder:

```
cd "$HOME\Documents"
git clone https://github.com/yositomi-opu/stack_questions.git
cd stack_questions
dir
```

Tip

Enter `explorer .` in PowerShell to open the cloned folder in File Explorer.

3. Run the first-time setup

1 Open the Windows launcher

In File Explorer, open **stack_questions**, then **scripts - windows**, and double-click **start-mcq-webapp.bat**.

2 Handle a Windows warning

If Windows displays a publisher warning, first confirm that the file is inside the repository you cloned. Follow your organization's security policy on managed computers.

3 Choose the STACK location

At the first **STACK location** prompt, press Enter without typing anything if you do not already have STACK. The launcher downloads moodle-qtype_stack automatically.

```
STACK code setup is required.
If STACK is not cloned yet, press Enter to download it automatically.
STACK location: [press Enter here]
```

Existing STACK clone

If you already have moodle-qtype_stack, enter its full folder path, for example
C:\work\moodle-qtype_stack.

4 Wait for the automatic check

The launcher checks Python, Maxima, and the STACK code. Setup is complete when **STACK code: OK** and the local CAS availability message appear.

```
Python: OK
Maxima: OK (...)
STACK code: OK
STACK loading mode: ...
The MCQ WebApp local CAS evaluation is available.
```

Normal fallback

If a dumped Maxima executable cannot be created, the launcher automatically switches to loading STACK at evaluation time. This is expected behavior.

4. Open the web app

1 Confirm browser startup

After setup, your default browser should open automatically. If it does not, open either URL manually:

```
http://127.0.0.1:4173/
http://localhost:4173/
```

2 Verify STACK evaluation

In Problem variables, enter an expression containing `rand([1, 2, 3])`, then click **Evaluate problem variables**. If 1, 2, or 3 appears instead of the unevaluated expression, STACK is working.

3 Stop the server

Select the Command Prompt window used by the launcher and press **Ctrl+C**. Closing that window also stops the server.

Troubleshooting

Symptom	What to do
Browser does not open	Open <code>http://localhost:4173/</code> manually.
Address already in use	The server is already running. Open <code>localhost:4173</code> .
Python is not found	Reinstall Python and enable Add <code>python.exe</code> to PATH.
Maxima is not found	Check Maxima. If necessary, set <code>MAXIMA_EXECUTABLE</code> to the full path of <code>maxima.bat</code> .
STACK code: not loaded	Run the launcher again and provide the STACK location, or press Enter for automatic download.
<code>rand(...)</code> remains unchanged	Restart the server and reload the browser page.

More help See `app/mcq-webapp/README.md`. In PowerShell, run `py -3 app\mcq-webapp\server.py --check` for diagnostics.